

Final Report

C. Yazan*, A. Hatem†

Abstract—This paper addresses a significant challenge in the field of Automatic Modulation Classification (AMC) — the degradation of classification accuracy under low Signal-to-Noise Ratio (SNR) conditions. Building upon insights gained from a comprehensive literature review, this study focuses on overcoming the limitations observed in existing AMC methodologies, particularly those related to the implementation of Early-Exiting strategies[3]. While Early-Exiting has proven effective in reducing computational time and maintaining accuracy for high SNR signals, it has been less effective for low SNR signals, leading to unnecessary computational overhead and suboptimal accuracy. To address this, we explore various approaches used in the literature for improving AMC performance in low SNR environments. We then propose two novel solutions: the first involves using distinct models for different SNR ranges combined with Early-Exiting techniques, and the second integrates selective denoising processes for lower SNR signals. Both proposals aim to enhance the AMC's robustness and accuracy in handling a wide range of SNR levels, particularly focusing on low SNR scenarios. Additionally, we address the challenges of developing some of these algorithms from scratch, considering the unavailability of pre-existing code. This paper contributes to the advancement of AMC systems, proposing innovative approaches that balance computational efficiency with improved accuracy across varying SNR conditions.

Index Terms—Automatic Modulation Classification, AMC, Deep Learning, DL, Early Exiting, EE, Multi Branch Neural Networks, Wireless, Signal-to-noise-ratio, SNR

I. INTRODUCTION

The importance of Automatic Modulation Classification (AMC) has grown significantly over the past decade in the field of wireless communication. AMC works to automatically classify a signal over a given radio spectrum. It facilitates a range of applications such as: interference detection and monitoring, dynamic spectrum access, and enhanced target recognition in radar systems. AMC is also of great interest to regulatory and defense organizations as it enables the detection of illegal and malicious spectrum usage. AMC is therefore of fundamental importance as beyond 5G and 6G applications focus on connectivity of critical infrastructure. This technology has been seen in various applications such as Software defined radios (SDR), military applications etc. AMC can be classified into two categories: featured-based (FB) methods and likelihood-based (LB) methods. [2]

In non-cooperative communication contexts, FB methods are favored over LB methods due to the lack of pre-existing knowledge and the complex computation required for likelihood ratio functions with uncertain parameters. LB modulation classifier works by comparing the likelihood function value of a received signal against a known set of modulations. This method has been effectively used for modulation classification in multi-channel environments, achieving high accuracy [4]. However, it requires certain parameters, like carrier frequency,

code rate, and channel parameters, to be known beforehand. The complexity of this method significantly increases when dealing with unknown parameters. FB methods function through a two-step process: firstly, extracting features by hand from the received signals, such as higher-order statistics, entropy characteristics, and power spectral density; and secondly, classifying based on these extracted features. Even though the approach is straightforward and requires minimal computation, it struggles with complex, non-linear issues [13]. Moreover, the manual selection of features might not accurately capture the diverse modulations in signals, and choosing inappropriate features can lead to lower accuracy in the classifier's ability to recognize and classify these signals. Over the last decade, deep learning (DL)-based automatic modulation classification (AMC) has emerged as a significant field, benefiting from the advancements in deep learning technologies. This method employs neural networks for automatic feature extraction, circumventing the necessity for manual feature engineering. Nonetheless, as DL models have evolved, their demands for power and computational resources have substantially increased. This presents a challenge for smaller-scale devices, including embedded systems and IoT devices.

The culmination of our latest literature review, conducted in collaboration with my research professor, has steered our focus towards a pivotal issue identified in the seminal paper [3]. This issue centers around the challenges faced by Automatic Modulation Classification (AMC) systems when operating under low Signal-to-Noise Ratio (SNR) conditions. While the concept of Early-Exiting has been instrumental in enhancing computational efficiency and maintaining accuracy in higher SNR environments, its application has paradoxically led to unnecessary computational overhead in lower SNR scenarios. In such cases, the model fails to exit early and continues to exhibit suboptimal accuracy levels.

Recognizing this critical gap, the primary objective of this paper is to delve into the various strategies employed by researchers to address low SNR challenges in AMC. Our aim is to not only understand and evaluate these methodologies but also to investigate the potential integration of Early-Exiting techniques within these frameworks. This exploration is motivated by the hypothesis that Early-Exiting, when applied judiciously, could further optimize the performance of AMC systems, especially in low SNR contexts.

Moreover, this paper endeavors to propose innovative solutions that can be seamlessly incorporated into the existing AMC framework discussed in the aforementioned paper. These proposed solutions are envisioned to enhance the model's resilience and accuracy in low SNR situations, thereby addressing a critical limitation in the current landscape of AMC technology. The technique that we wish to test is incorporating

a denoiser in order to help improve the results of lower SNR signals. Through this research, we aspire to contribute meaningful advancements to the field of AMC, particularly in optimizing its functionality across varying SNR ranges.

II. RELATED-WORK

Over the last decade, there has been a substantial body of research dedicated to the advancement of Adaptive Modulation and Coding (AMC) technology, particularly in the domains of signal processing and communication. This research has yielded intelligent solutions for the effective oversight and supervision of radio communication spectra. In the realm of wireless communications, there has been a shift in focus to develop an AMC model characterized by low complexity and robustness, with a particular emphasis on enhancing accuracy in scenarios featuring low Signal-to-Noise Ratio (SNR) conditions. This section focuses on investigating research papers to understand different approaches to improve the classification accuracy of low SNR input signals, as well as the paper created by Hatems research group in order to understand how to incorporate it with low-SNR solutions.

Robust Automatic Modulation Classification in Low Signal to Noise Ratio

[1]

This paper [1] investigates enhancing AMC efficiency in low SNR environments. The research introduces a threshold autoencoder denoiser convolutional neural network (TAD-CNN) that pre-processes input signals through a noise-reducing auto-learning sub-network. This preparation stage is crucial for improving signal classification accuracy by a subsequent CNN, especially under challenging low SNR conditions. The study's significance lies in its potential to substantially advance AMC performance where noise levels are typically problematic.

Basic Components of CNN for AMC

CNN Architecture: CNNs are deep learning models composed of multiple layers that process multi-dimensional inputs, like 2D images. These networks mimic certain aspects of human visual perception and are highly efficient in parameter usage compared to fully connected networks. They excel at tasks like image and signal classification due to their deep, hierarchical structure.

Denoise Layer: The denoise layer leverages an Autoencoder to clean the input signals by eliminating unwanted noise, preparing cleaner data for the subsequent layers of the CNN.

Activation Functions

The CNN utilizes various activation functions to introduce non-linearity into the model, which helps in learning complex patterns:

- 1) **ReLU (Rectified Linear Unit):** Provides a simple, efficient way to introduce non-linearity, turning off negative values to avoid vanishing gradient issues.
- 2) **SeLU (Scaled Exponential Linear Unit):** Adjusts the scale of outputs, aiming to normalize the activations to

maintain a mean of zero and variance of one during training.

- 3) **Softmax:** Typically used in the final layer of a classification network, it converts logits to probabilities by comparing the exponential score of each class against the sum of exponential scores for all classes.
- 4) **Sigmoid:** Squeezes the output into a range between 0 and 1, useful for binary classification.

Convolutional Layers

These layers are the core building blocks of a CNN, where each layer filters the input data with learnable kernels to create feature maps. The output of a convolutional layer can be expressed as:

$$y_j = \sum_{i \in S} x_i * k_{ij} + b_j$$

where x_i corresponds to the i^{th} feature map, y_j is the output for the j^{th} channel, k_{ij} represents the kernel, and b_j is the bias.

Batch Normalization (BN)

BN is a technique to standardize the inputs of each layer within the network, aiming to stabilize learning by reducing internal covariate shift. The normalized value $\hat{x}_i$ is calculated using the mean μ and variance σ^2 of the current mini-batch:

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

Pooling Layers

Pooling layers downsample the spatial dimensions of the input feature maps, reducing the number of parameters and computations, which helps to control overfitting and allows the network to learn higher-order features.

Dropout Layer

The dropout layer is used to prevent overfitting by randomly deactivating a subset of neurons during training, which encourages the network to learn more robust features.

Integration into AMC

The AMC system uses these components to process and classify modulation schemes effectively. The combination of denoising, convolutional processing with various activation functions, batch normalization, pooling, and dropout contributes to a powerful system capable of distinguishing between different types of modulated signals, even in the presence of noise.

This AMC system is designed to leverage the strengths of CNNs for signal classification tasks, making it a suitable candidate for robust and efficient modulation classification in noisy environments, as encountered in real-world communication systems.

Threshold Autoencoder Denoiser CNN (TAD-CNN)

Threshold Autoencoder Denoiser (TAD)

The TAD is a sophisticated component of the TAD-CNN that includes a Batch Normalization (BN) layer and an autoencoder architecture. The BN layer normalizes the input data to stabilize learning and improve generalization. The autoencoder part, comprising an encoder and a decoder, is designed to denoise the input signals.

Basic Components of TAD-CNN

- **Convolutional Layers:** Utilize kernels to extract features from input signals, creating feature maps that retain important spatial information for classification.
- **Activation Functions:** Non-linear functions like ReLU, SeLU, Sigmoid, and Softmax introduce non-linearity into the model, allowing it to learn complex patterns from the data.
- **Batch Normalization:** Standardizes the outputs of neurons to ensure stable distribution of activations throughout the network.
- **Pooling Layers:** Reduce the spatial dimensions of the feature maps, which helps to decrease the computational load and control overfitting.
- **Dropout Layer:** Randomly deactivates neurons during training to prevent overfitting and encourage the network to learn more robust features.

The Denoise Algorithm

It utilizes a thresholding technique to remove noise from the signals, setting values below a certain threshold to zero. This process is aimed at ensuring that only the most significant features are preserved, enhancing the AMC capability of the network.

TAD-CNN Network Structure

- **TAD Denoiser:** Aims to reduce noise in the input signal, enhancing the quality of the features fed into the CNN.
- **CNN Part:** Comprises convolutional layers and ResNet blocks, where ResNet blocks help prevent vanishing gradient problems by utilizing skip connections.
- **Fully Connected Part:** Includes dense layers that classify the modulation scheme from the denoised and processed features.

Principles of Denoise

- **Threshold Computation:** The TAD applies a computed threshold to the features, turning less significant values into zeros, effectively denoising the signal.
- **Signal Transformation:** Transforms the input signals into a domain where useful information can be extracted, and noise is minimized.

ResNet Block Architecture

It incorporates convolutional layers with skip connections, ensuring that the network can learn deeper features without the degradation of the learning signal.

Optimization and Loss Function

- **Early Stopping and Alpha Dropout:** Techniques used to prevent overfitting, where Early Stopping terminates training if there is no improvement, and Alpha Dropout maintains the mean and variance of the inputs during the dropout.
- **Adam Optimizer:** Utilized for its efficiency in updating network weights.
- **Cross-Entropy Loss:** Employed as the loss function, it calculates the difference between the predicted probabilities and the actual class labels.

Simulation Results

- **Accuracy:** TAD-CNN shows a high level of accuracy in AMC, particularly at both low and high SNR levels, due to its robust denoising and classification capabilities.
- **Comparison with Other Methods:** The TAD-CNN is compared against other methods and found to provide better classification performance, especially in the presence of noise.

The TAD-CNN architecture is a comprehensive approach to AMC that improves signal quality through denoising and leverages deep learning techniques to classify modulation schemes effectively. The network's design incorporates modern neural network techniques to deal with the challenges of noisy signals, making it highly relevant for research in signal processing and communications.

Implementation Details

- **Dataset:** The RadioML 2016.10A dataset is used, a widely accepted benchmark in the academic community for AMC, containing various modulated signals at SNR levels from -20dB to 18dB.
- **Algorithm:** A proposed TAD-CNN method is outlined, including steps for dataset division, model initialization, hyperparameter setting, training, and validation.

Denoise Performance Analysis

- **TAD:** A denoise function is proposed within the TAD to improve classification accuracy by cleaning raw signals, particularly in low SNR conditions.
- **Adjustment Parameter:** An adjustment parameter (ρ) influences the effect of the denoise process on classification accuracy.

TAD-CNN Network Structure

- **Denoiser:** The TAD-CNN includes a denoising layer designed to reduce noise in the input signal, which helps to improve classification accuracy in the AMC task.
- **CNN Architecture:** The network comprises convolutional layers and ResNet blocks, with the latter featuring skip connections to aid in deeper learning.

Results

- **Classification Accuracy:** The TAD-CNN model demonstrates significant improvement in classification accuracy over models without the denoiser, especially in low SNR situations.
- **Comparison with Other Models:** The TAD-CNN shows a higher recognition accuracy compared to other architectures such as MC-CNN, CNN/VGG, and RN (Residual Network), as highlighted in the comparison table.

Conclusion Drawn from the Results

- **Enhanced Accuracy:** The TAD significantly enhances the performance of AMC models in both high and low SNR situations.
- **Low SNR Performance:** The denoiser's impact is particularly notable in low SNR scenarios, where it helps the model achieve a high level of accuracy, proving the denoiser's effectiveness.
- **High SNR Performance:** Even at high SNR levels, the TAD-CNN improves classification accuracy, illustrating the denoiser's overall benefit to the AMC model.

The TAD-CNN's architecture successfully addresses the challenge of noise in modulation classification, providing a robust solution that enhances the clarity and accuracy of signal classification across a wide range of SNR levels. This approach is especially critical in real-world scenarios where signal noise can significantly hinder the performance of AMC systems.

Using Early Exits for Fast Inference in Automatic Modulation Classification

Baseline Model AMC

The baseline model for AMC in the paper is a 1D-CNN with six convolutional layers that feature ReLU activation and batch normalization, aiming for efficient feature extraction from input signals. It includes a max-pooling layer for dimensionality reduction, and three fully connected layers ending in a SoftMax output layer. This structure takes advantage of CNNs' ability to handle parameter sharing and translational invariance, making it adept at recognizing hierarchical signal patterns necessary for modulation classification. Dropout layers are also used to prevent overfitting, ensuring robustness in the model's predictive capability.

Proposed Early Exiting Architectures

The proposed early-exiting architecture for AMC is designed to allow a neural network to make classifications at intermediate points before completing the full sequence of layers. This system includes multiple exits at different stages, enabling the model to output a classification when it reaches a sufficient level of confidence, thereby potentially reducing computation time. The critical aspects of this design are the placement of these exit points and the determination of a confidence threshold, which is often based on the entropy of the predictions, to decide if the inference process can be stopped early. The goal is to strike a balance between quick inference and maintaining high accuracy, optimizing

for different signal-to-noise ratios. The training algorithm has been adapted to support this structure, incorporating separate loss calculations and parameter updates for the early exit paths and the main network path.

Early Exit Training Engine

The early-exit training engine for the AMC architecture incorporates a training pass that consists of two forward passes. The first pass computes outputs using common layers shared by both the main and early-exit branches, while the second pass completes the remaining layers of the main backbone network. Two separate losses are calculated: one for the early-exit output (Loss 1) and another for the main network's output (Loss 2). Loss 1 is used for updating the shared layers, while Loss 2 is used to update the layers beyond the early-exit branch. This strategy enables efficient learning for both early-exit and full-path classification, optimizing the network to make accurate predictions whether exiting early or at the final layer.

Early Exit Inference Engine

The early-exit inference process in the AMC model involves computing the output of a network sample and determining if it can conclude early based on entropy evaluation. This evaluation measures the confidence level of the classification—if it's high enough, it suggests that further processing may not be necessary. The entropy, a measure of prediction uncertainty, is calculated for the sample's output, and if it's below a set threshold, the sample is classified immediately. Otherwise, it proceeds to the second forward pass for further processing. This process effectively balances speed and accuracy in classification tasks. The study's results indicate that signals with low SNR suffer from reduced accuracy, making them less amenable to the benefits of early-exit architectures. The low SNR leads to higher uncertainty in early layers, necessitating deeper processing to achieve accurate classification. Thus, early exits are less likely to be utilized for low SNR signals, as these signals require more comprehensive analysis to reach the same level of confidence and accuracy as high SNR signals.

Results

The results from the study reveal that the effectiveness of early exits in AMC is influenced by the signal-to-noise ratio (SNR) and the exit threshold. Early exits close to the input can classify high SNR signals quickly and accurately. However, as SNR decreases, more computational depth (later exits) is required for accurate classification, increasing inference time. Trade-offs between accuracy and speed are managed by adjusting the entropy threshold for early exits, suggesting an adaptive approach based on SNR might be most efficient. These insights emphasize the need for strategic placement of early exits and dynamic thresholding in the design of neural networks for AMC.

III. PROPOSAL

Informed by the latest literature review and aiming to refine SNR challenges as discussed in Hatem’s research, we propose integrating a TAD-denoiser into the early-exit AMC framework. The rationale is that an effective denoiser could enhance the precision of early exits, thereby streamlining the AMC process and potentially reducing overall inference latency. It is crucial, however, to consider the trade-off introduced by the denoiser’s computational demand. High SNR signals may not necessitate denoising, which implies that the architecture should adaptively deploy the denoiser, optimizing the balance between enhanced accuracy and expedited inference.

A. Design

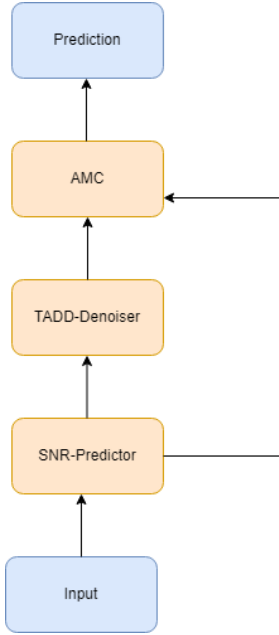


Fig. 1. High-Level Design of AMC with a TAD-Denoiser

Figure 1 outlines our engineered approach. It starts with the input signal being assessed by a simple CNN SNR-Predictor designed to efficiently estimate the signal’s SNR. Should the SNR fall below a specific threshold, the signal is routed through the TAD-Denoiser for noise reduction. Conversely, signals with an SNR above the threshold bypass this step and proceed directly to AMC. This design ensures high SNR signals bypass unnecessary processing, thereby optimizing inference times for these signals while allocating denoising resources to those signals that stand to benefit from it.

Signals with low SNR are directed through the TAD denoiser suggested in [1], enhancing their clarity before classification. The AMC then processes the refined signal, ensuring that the system’s resources are effectively allocated to signals that genuinely require denoising. This selective approach aims to improve the accuracy and efficiency of the classification process.

B. Execution

Prior to adoption, we’re setting out to validate the necessity of the TAD-Denoiser. By harnessing the TAD-Denoiser and CNN framework from [1], we’ve run trials across varied datasets to scrutinize its efficacy. Our comparative analysis involves performance metrics both with and without the denoiser in the loop. To deepen our insights, we’ve employed a stratified k-fold strategy for training, offering a robust basis for evaluation and an enriched comprehension of the denoiser’s impact. The datasets used are the RML-2016a, RML-2016b and RML-2022 dataset (which is the one used in the paper at hand).

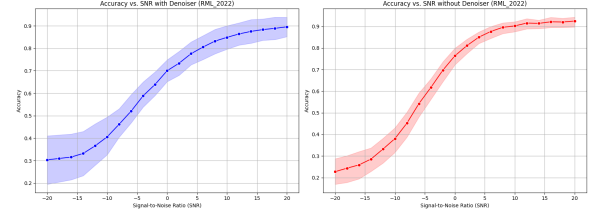


Fig. 2. Accuracy vs SNR RML 2022

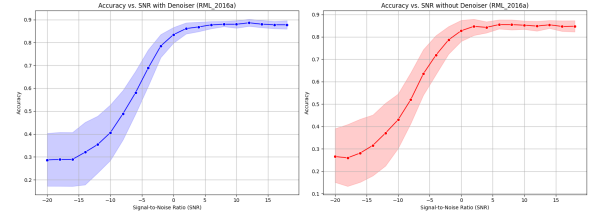


Fig. 3. Accuracy vs SNR RML 2016a

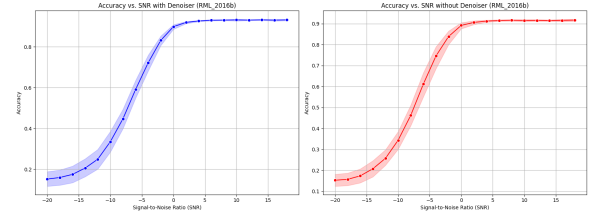


Fig. 4. Accuracy vs SNR RML 2016b

Figures 2, 3, and 4 offer a comparative analysis, illustrating that while the TAD-denoiser enhances accuracy for signals with SNRs below zero, the absence of the denoiser yields better results for higher SNR signals. This pattern suggests that while the TAD-CNN’s hyperparameters are tuned to eliminate noise, they might also inadvertently dampen the actual signal, leading to the diminished performance where the signal is inherently strong. This further reinforces the need for SNR-Predictor that is able to approximate the SNR value of the signal and if it is above a certain threshold then the TAD Denoiser is not necessary.

Next we will evaluate a box-plot comparing model of a given dataset with the denoiser and without the denoiser

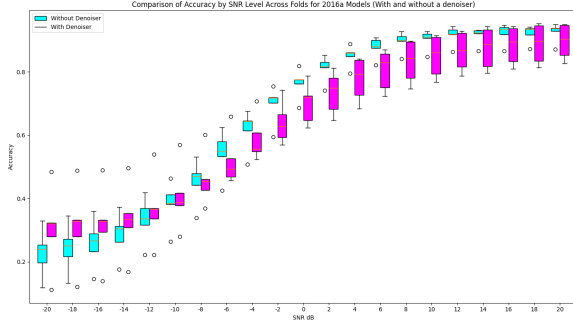


Fig. 5. Accuracy vs SNR RML 2022

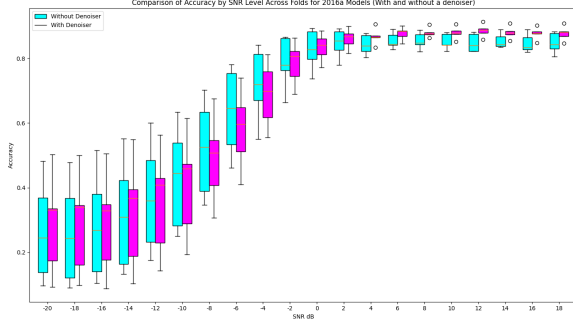


Fig. 6. Accuracy vs SNR RML 2016a

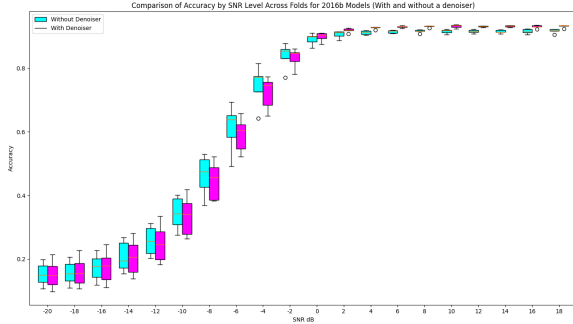


Fig. 7. Accuracy vs SNR RML 2016b

An analysis of Figures 5, 6, and 7 indicates a lower variability in the performance of models using the TAD-denoiser compared to those without, alongside improved accuracy, suggesting more reliable outcomes. The exception noted is in the 2022 dataset, where higher SNR levels correspond to increased variance with the TAD-denoiser. To address this, it's crucial to configure the system to bypass the denoising process for higher SNR signals, thereby aligning the variance with expected performance.

C. Designing the SNR Predictor

The objective is to develop a streamlined, low-power neural network capable of estimating the SNR of incoming signals with reasonable accuracy. This simplicity is crucial to ensure the model can operate on very small-scale devices without excessive power consumption or memory demands, avoiding

the complexity typically associated with more robust models. This approach aims to strike a balance between functional accuracy and operational efficiency, suitable for deployment in resource-constrained environments.

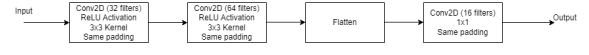


Fig. 8. Initial SNR predictor design

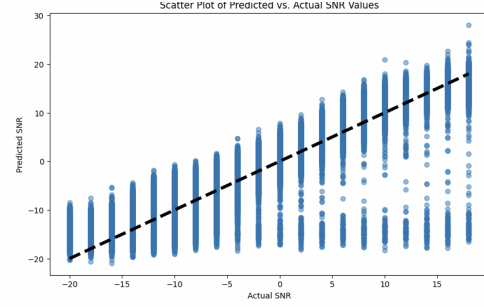


Fig. 9. Initial SNR predictor design

The model was developed using the RML 2016b dataset, employing a stratified K-fold approach to ensure robust cross-validation. Mean Average Error (MAE) was utilized as the evaluation metric, appropriate for this regression task which predicts a single continuous output. This method underscores our commitment to achieving accurate and reliable predictive performance across diverse signal conditions.

To interpret the results in Figure 9 we must consider the following:

- **Concentration Along the Line:** Ideally, data points should cluster near the dashed line, showing that predictions closely match the actual values.
- **Vertical Spread:** Points spread vertically at different actual SNR values indicate errors; points above the line are overestimations, those below are underestimations.
- **Density of Points:** High density of points far from the line suggests consistent prediction errors, indicating systematic inaccuracies.
- **Symmetry:** Symmetric distribution around the line suggests unbiased errors. Asymmetry indicates a tendency to overpredict or underpredict.
- **Extreme Values:** Points distant from the line at the extremes of the x-axis highlight struggles with very high or low SNR values.

The analysis indicates a potential quantization effect in predictions, with significant variability in predictions across SNR levels. This spread of predictions suggests a need for further model refinement to enhance accuracy across all conditions.

Following the initial analysis, the model architecture was refined to enhance feature extraction capabilities, particularly for characterizing SNR values more effectively. Additionally, to address the underperformance at extreme SNRs, the model's SNR predictor was trained using a broader range of data,

including the RML 2016a, 2016b, and 2022 datasets. This approach aimed to enrich the model with diverse scenarios and edge cases, thereby improving its robustness and accuracy. With this in mind, the goal is to also ensure that the model doesn't overfit or underfit to the data.

The neural network structure designed for Signal-to-Noise Ratio (SNR) prediction is tailored to effectively capture the complex patterns and relationships indicative of signal quality. The model architecture facilitates accurate SNR prediction through several components:

1) **Input Layer:**

- *Purpose:* Takes in raw or preprocessed signal data, which could be time-series samples or transformed data (e.g., spectral components).
- *Effectiveness:* Allows the model to learn directly from the fundamental characteristics of the signal crucial for assessing SNR.

2) **Dense Layers (Hidden Layers):**

- *Purpose:* These layers are where the actual learning happens, with each neuron potentially representing different aspects or features of the input data.
- *Effectiveness:*
 - a) The first dense layer acts as a feature extractor, transforming the raw input into a space where features indicative of SNR become distinguishable.
 - b) The second dense layer further refines these features, allowing the network to build complex representations that correlate effectively with SNR.

3) **Dropout Layers:**

- *Purpose:* Randomly deactivates a percentage of neurons during training to prevent the model from becoming overly reliant on any single neuron or pathway, thus reducing overfitting.
- *Effectiveness:* Ensures that the network generalizes well to new, unseen data by promoting robustness, crucial for applications where signal conditions may vary.

4) **Output Layer:**

- *Purpose:* Consolidates the learned features into a single continuous value representing the predicted SNR.
- *Effectiveness:* Provides a straightforward quantitative output that directly corresponds to SNR, facilitating interpretation.

5) **Loss Function and Optimization:**

- *Purpose:* Uses Mean Squared Error (MSE) for the loss function and Adam optimizer for learning. MSE is effective for regression problems as it emphasizes larger errors and aggressively minimizes them.
- *Effectiveness:* Adam optimizer is advantageous due to its adaptive learning rate capabilities, helping the model converge faster to the minimum loss in complex and noisy data environments.

Overall, this model structure is adept at learning a hierarchy of features, from the most basic to the most intricate, and directly correlating them to SNR, which is essential for precise prediction in communication systems.

D. Analysis of Model Performance

1) *Scatter Plot Analysis:* The scatter plot demonstrates a reasonably close correlation between predicted and actual SNR values, with many data points gathering near the diagonal line indicating perfect predictions. Notably, there are significant variances at extremes of the SNR scale where predictions substantially deviate from actual values.

2) *MAE per Fold Analysis:* The line graph of MAE per fold shows a rising trend in MAE with advancing folds, which may suggest instability in the model or inconsistencies in data splitting or model initialization.

3) *Mean MAE per SNR Value Analysis:* Analysis from the bar chart of mean MAE per SNR level reveals variable errors across different SNR conditions. The model exhibits improved accuracy at moderate SNR values and poorer performance at extreme SNRs, highlighting challenges in managing very low or high SNR scenarios.

4) *Overall Assessment:* While the model generally captures the trend of SNR values effectively, as indicated in the scatter plot, there is significant potential for enhancing consistency across folds and better managing extreme SNR values.

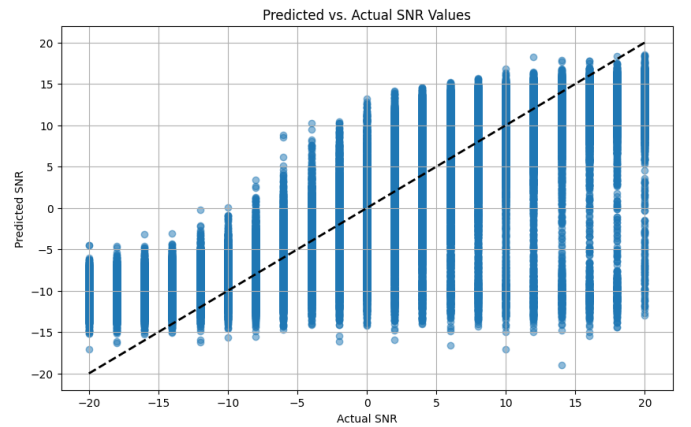


Fig. 10. Updated SNR predictor design results

The analysis of model performance across different folds highlighted that the model trained with the first fold (K-Fold = 1) exhibited the smallest Mean Absolute Error (MAE) compared to all other folds. This metric indicates that the model from the first fold achieved the highest accuracy in predicting the target variables, making it the most reliable among the iterations tested.

Given its superior performance, this model was selected for further development. It has been saved and stored for integration into the TAD-CNN. The choice to proceed with this specific model iteration is based on its demonstrated predictive

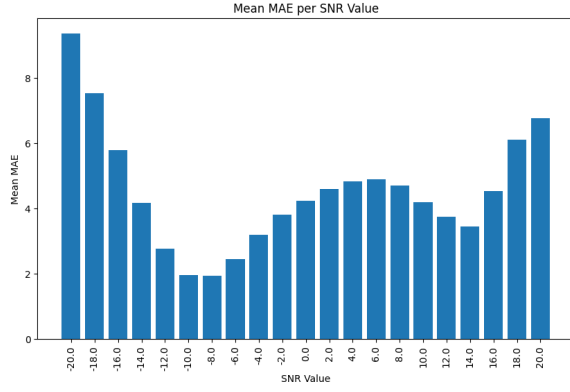


Fig. 11. MAE SNR Predictor

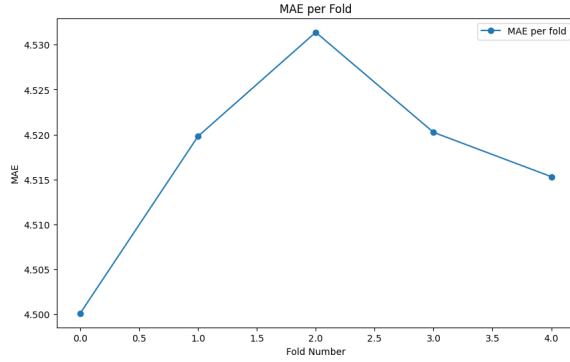


Fig. 12. MAE of Fold SNR Predictor

strength and stability, which are crucial for the robustness and effectiveness of the TAD-CNN in operational settings.

The model architecture, including the SNR predictor, was deliberately designed to balance performance with computational efficiency. While there is potential for a more complex SNR predictor that might yield marginally better performance under certain conditions, the need for deployment on a small-scale device necessitates a more streamlined approach. This consideration ensures that the entire architecture can be deployed effectively without overburdening the device's computational resources, and at the same time reducing the inference time.

The selection process involved careful consideration of the model's error metrics and generalization capabilities across diverse scenarios, as reflected in the cross-validation procedure. The deployment of this model into the full system is anticipated to enhance the overall performance and reliability of the TAD-CNN, providing improved analytical capabilities in real-world applications. This approach aligns with the project's objectives to deploy an effective yet resource-efficient solution suitable for small-scale devices.

E. Putting it all together

The subsequent phase involves the integration of the SNR Predictor into the broader Traffic Analysis and Detection Convolutional Neural Network (TAD-CNN) architecture. As

illustrated in Figure 1, the SNR Predictor is strategically positioned immediately following the input layer. This placement allows it to efficiently assess the incoming signal and determine its Signal-to-Noise Ratio (SNR).

The operational logic of the SNR Predictor is as follows:

- **SNR Estimation:** Upon receiving the input, the SNR Predictor computes an estimated SNR value for the signal.
- **Conditional Processing:**
 - **Low SNR ($\leq$ Threshold):** If the estimated SNR value is less than or equal to a predefined threshold, indicative of significant noise, the signal is routed through the TAD-Denoiser. This step is crucial for enhancing signal clarity before further processing.
 - **High SNR ($>$ Threshold):** If the estimated SNR exceeds the threshold, indicating minimal noise interference, the signal bypasses the TAD-Denoiser and is directly fed into the Automatic Modulation Classification (AMC) network. This approach reduces computational overhead by avoiding unnecessary denoising of already clear signals.

This conditional processing not only optimizes the network's efficiency but also ensures that the computational resources are judiciously used, enhancing the system's overall performance in real-time applications. The integration strategy described ensures that the network remains robust and responsive across varying signal conditions, adhering to the overarching goal of deploying an effective yet resource-efficient solution in small-scale devices.

The expected outcomes from integrating the TAD-Denoiser within the network are pivotal in assessing the functionality of the SNR Predictor. Specifically, the results when the TAD-Denoiser is utilized should align closely with those observed at the SNR threshold range. Ideally, the performance metrics at this threshold should either match or be very close to each other. This alignment will serve as a critical indicator of the SNR Predictor's operational efficacy.

The implications of the outcomes are as follows:

- **Consistency at Threshold:** Consistent results at the SNR threshold would validate the SNR Predictor's accuracy in classifying signal noise levels correctly.
- **Variance Implications:** A higher variance in the results might suggest potential issues with the SNR Predictor. Specifically, it could indicate that the SNR Predictor is misclassifying some signals. As a consequence, these misclassified signals could either erroneously bypass or be unnecessarily processed by the TAD-Denoiser, which might skew the overall results.
- **Error Identification:** Conversely, if the variance is low and the results are consistent across the threshold, it implies robust signal classification and effective denoising, affirming the system's design and functional intent.

These observations will guide further refinements to the SNR Predictor, aiming to minimize errors in signal classification and enhance the overall accuracy and reliability of the

TAD-CNN architecture.

1) *Analyzing the results:* Following the integration of the SNR-Predictor into the network, comprehensive evaluations revealed that the optimal threshold for SNR is 5 dB. This finding is crucial for tuning the network's responsiveness to varying noise levels in signal processing.

The model underwent extensive training using datasets from RML 2016a, 2016b, and 2022. This diverse training regimen was strategically chosen to ensure that the results could be effectively compared before and after the inclusion of the SNR Predictor. The objective was to ascertain not only the improvements in signal processing accuracy but also to gauge the added value brought by the SNR Predictor in practical scenarios.

For validation:

- **Stratified K-Fold Cross-Validation:** A stratified k-fold cross-validation approach was employed to determine the best performing model iteration and to assess the robustness of the system. This method ensured that each fold of the dataset was representative of the whole, thereby maintaining the integrity and diversity of the dataset across training and validation phases.
- **Comparative Analysis:** By comparing performance metrics before and after integrating the SNR Predictor, we were able to quantify the enhancements in model accuracy and system efficiency.

These steps are instrumental in reinforcing the system's robustness and ensuring that the SNR Predictor contributes positively towards the overall goal of achieving high accuracy in real-world operational conditions.

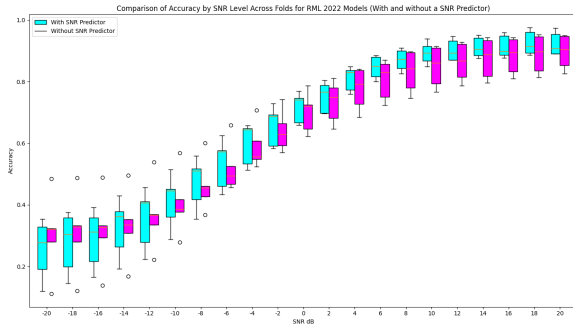


Fig. 13. Accuracy vs SNR RML 2022

Tying the detailed analysis of the SNR predictor's performance from the MAE distribution into the broader context provided by the box plots across different datasets shown in Figures 13, 14 and 15, we can more deeply understand the operational impacts of integrating an SNR Predictor within the signal processing framework. Here's a cohesive synthesis of the insights:

1) Performance at Low SNR Values:

- The higher MAE at low SNR values seen in the bar chart correlates with the increased variance observed in the box plots for models with an SNR Predictor.

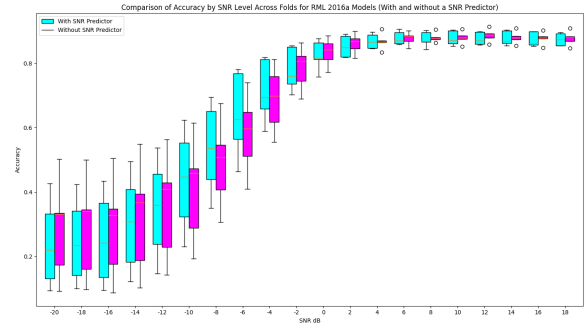


Fig. 14. Accuracy vs SNR RML 2016a

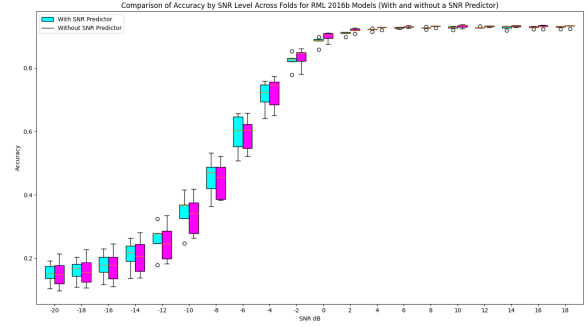


Fig. 15. Accuracy vs SNR RML 2016b

This suggests that inaccuracies in SNR prediction at low levels might lead to inappropriate activation or bypass of the TAD-Denoiser, thereby affecting the overall accuracy negatively.

2) Optimal Performance at Moderate to High SNR Values:

- The lower MAE in the moderate SNR range where the predictor performs optimally supports the observations from the box plots, which indicate improved median accuracy and reduced variance in models with the SNR Predictor at higher SNR levels. This alignment suggests that the predictor effectively discerns when to apply or skip denoising processes, enhancing model reliability and signal fidelity in environments with better signal-to-noise ratios.

3) Impact on System Design and Deployment:

- The insights into the predictor's performance help in pinpointing areas for model tuning, particularly for enhancing prediction accuracy at extreme SNR values. This is critical for applications requiring robust performance across a wide range of conditions, such as communications in variable noise environments.

4) Dataset Variability and Adaptability:

- The performance trends across the RML 2016a, 2016b, and 2022 datasets further emphasize the SNR Predictor's adaptability to different signal characteristics. This adaptability is crucial for deploying systems that must perform consistently across di-

verse operational scenarios and signal conditions.

In essence, while the SNR Predictor significantly enhances system performance in clearer signal conditions, the challenges at lower SNR levels highlight the need for continuous improvement of the predictor's accuracy and decision-making capabilities. Refining the predictor's threshold settings and decision algorithms based on comprehensive performance analysis like this will ensure that it more reliably enhances system accuracy, especially in complex noise environments.

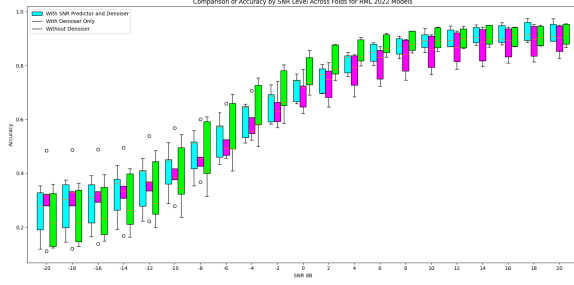


Fig. 16. Accuracy vs SNR RML 2022

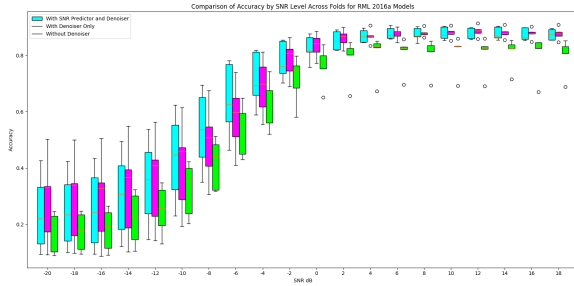


Fig. 17. Accuracy vs SNR RML 2016a

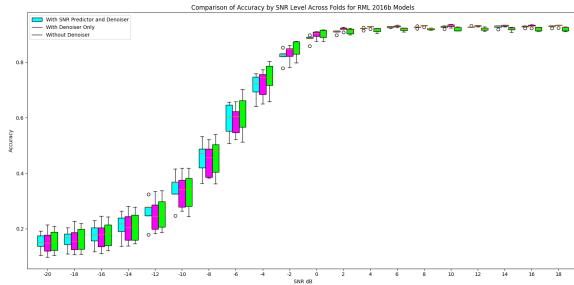


Fig. 18. Accuracy vs SNR RML 2016b

The box plots presented in Figures 16-18 show a comparison of model performance at different Signal-to-Noise Ratio (SNR) levels across folds for RML 2016a, 2016b, and 2022 models, with three configurations: with both SNR Predictor and Denoiser, with Denoiser only, and without Denoiser.

Analysis of the Results:

1) Higher SNR Levels:

- **With SNR Predictor and Denoiser:** Models consistently perform better as the SNR level increases.

This suggests that the SNR Predictor, when combined with the denoiser, is effectively identifying cleaner signals and enhancing the denoising process or bypassing it when not needed.

- **With Denoiser Only:** These models also show improvement with increased SNR but generally have lower accuracy and higher variability than the models that include the SNR Predictor. This indicates that the denoiser alone might sometimes be overly aggressive, potentially distorting signals that are already of good quality.
- **Without Denoiser:** This setup shows the least variance and steady performance improvement as SNR increases. This configuration benefits purely from better signal quality without any alterations from denoising processes, which could otherwise introduce errors in already clear signals.

2) Lower SNR Levels:

- **With SNR Predictor and Denoiser:** There is significant variability in performance, which could be due to the SNR Predictor's challenges in accurately classifying very noisy signals and deciding when to apply the denoiser.
- **With Denoiser Only:** Performance is generally lower, reflecting the potential drawback of denoising when it might not be accurately tuned to the noise characteristics of the signal.
- **Without Denoiser:** Consistently lower performance at very low SNR values, as expected, since there is no mechanism to improve signal clarity through denoising.

3) Overall Insights:

- The integration of the SNR Predictor with the Denoiser generally provides the best outcomes at higher SNRs, optimizing performance by intelligently managing denoising.
- At lower SNRs, all models struggle, but the variability in the "With SNR Predictor and Denoiser" setup suggests that the SNR Predictor does influence outcomes by its decision-making on denoising application.
- The model without any denoiser performs with more stability across the SNR spectrum but fails to capitalize on potential improvements that denoising could provide at low SNR values.

Conclusion: These results underline the complexity of signal processing at varying noise levels and the importance of intelligent systems like SNR Predictors in distinguishing when and how to apply further signal processing steps such as denoising. The decision-making process of the SNR Predictor needs to be highly accurate, especially in lower SNR conditions, to truly benefit the overall system performance. Continuous refinement of the SNR Predictor's algorithms is essential to enhance its utility, particularly in challenging noise environments.

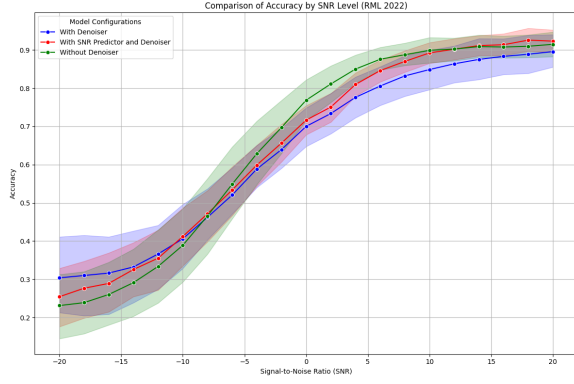


Fig. 19. Accuracy vs SNR RML 2022

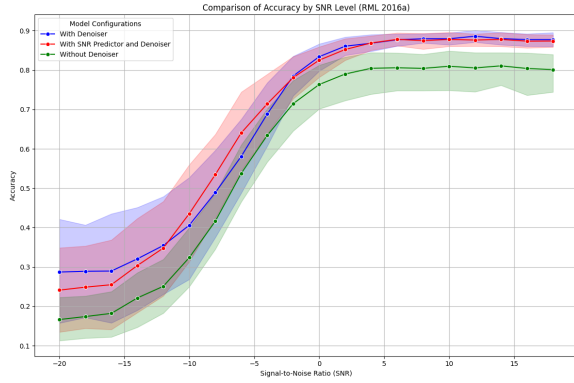


Fig. 20. Accuracy vs SNR RML 2016a

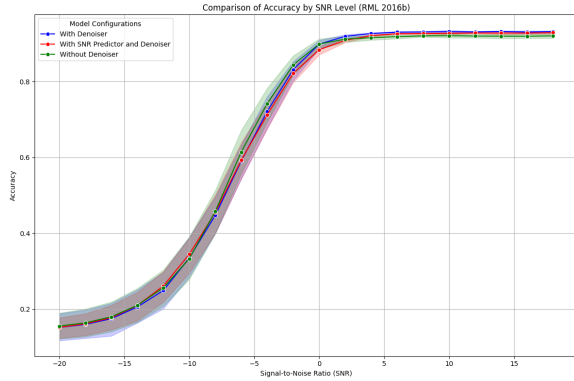


Fig. 21. Accuracy vs SNR RML 2016b

The series of plots in Figures 19-21 show the accuracy of radio modulation classification across different signal-to-noise ratio (SNR) levels for RML datasets from the years 2016a, 2016b, and 2022. These plots compare three different model configurations: using a denoiser, using both a SNR Predictor and a denoiser, and without using a denoiser.

OBSERVATIONS ACROSS ALL DATASETS:

- 1) **Performance Consistency:** All configurations tend to converge in performance at higher SNR levels (>0 dB), indicating that at high SNR, the clarity of the signal

reduces the necessity of either the denoiser or SNR Predictor.

- 2) **Impact of the SNR Predictor:** The inclusion of the SNR Predictor generally enhances performance, particularly in mid to high SNR ranges. This suggests that the SNR Predictor, when combined with the denoiser, effectively improves signal clarity or correctly decides when denoising is not necessary, thus preserving the original signal integrity.
- 3) **Utility of Denoiser Alone:** Models with only the denoiser and without the SNR Predictor show competitive performance, especially at lower SNRs. This indicates the inherent value of the denoising process under poor signal conditions. However, the performance gain from adding the SNR Predictor suggests that it helps optimize the denoising process by selectively activating it.
- 4) **Performance without Denoiser:** Models without any denoising mechanism underperform in lower SNR scenarios compared to the ones with denoising capabilities. This underlines the denoiser's role in improving signal readability at lower SNRs. However, the gap between models narrows at higher SNRs, emphasizing less dependency on denoising as signal quality improves.

SPECIFIC INSIGHTS:

- **RML 2016a & 2016b:** These datasets show a notable spread in performance between the configurations at lower SNR values, highlighting the challenges in handling noise. The SNR Predictor combined with the denoiser seems particularly effective in these cases, improving accuracy where the signal is heavily corrupted.
- **RML 2022:** The performance improvements are less dramatic but still evident, suggesting either a difference in signal characteristics in the dataset or improvements in the underlying model architecture over time.

Overall, these results validate the integration of SNR Predictors and denoisers in communication systems, especially for adaptive signal processing where conditions can vary widely. The SNR Predictor not only enhances the effectiveness of the denoising process but also ensures that system resources are optimally used by preventing unnecessary denoising operations. This adaptability is crucial for real-world applications where computational resources and energy efficiency are concerns.

IV. DISCUSSION

The research presented in this paper aimed to demonstrate the efficacy of incorporating a denoiser within Automatic Modulation Classifiers (AMCs) to significantly enhance accuracy, particularly at lower SNR levels. The results affirmatively indicate that the denoiser does enhance model performance where noise levels are high, a critical finding given the challenging nature of such environments. Further integration of an SNR predictor not only preserved accuracy at higher

SNRs but also optimized the system by preventing unnecessary processing, thereby conserving computational resources without compromising inference speed.

A. Potential Improvements

While the current system design shows promising results, there are several avenues for enhancement:

- **SNR Predictor Accuracy:** Improving the accuracy of the SNR predictor is paramount. However, this improvement must be balanced against the complexity of the predictor to avoid undue computational overhead. A more accurate but resource-intensive predictor might not be feasible for deployment on smaller devices.
- **TAD-Denoiser Optimization:** The TAD-Denoiser could be refined to be less computationally demanding while still effective, which would improve overall inference time and reduce power consumption.
- **Early Exiting for SNR Prediction:** Applying early exiting techniques directly to the SNR predictor could further accelerate the processing for high-SNR signals, where simpler signal processing paths might suffice.

B. Future Directions

Integrating the early exit architecture mentioned in previous studies [3] with the current model to test and refine these concepts further will be an essential next step. This integration could potentially create a highly efficient system capable of robust performance across a wide range of SNR scenarios. Testing under varied and real-world conditions will provide deeper insights into the practical viability and areas for refinement.

In conclusion, the enhancements through denoising and SNR prediction not only fortify the robustness of AMCs in adverse SNR conditions but also present a scalable approach to managing computational resources effectively in signal processing applications.

V. CONCLUSION

This study has contributed significantly to the field of signal processing by advancing the development and optimization of denoisers within Automatic Modulation Classifiers (AMCs). By integrating a refined denoiser alongside an SNR predictor, this paper has demonstrated marked improvements in AMC accuracy, particularly at lower SNRs where noise traditionally impairs signal integrity and classification reliability. The enhancements not only elevate the robustness of AMCs in challenging noise environments but also optimize computational efficiency, making the system suitable for deployment on devices with limited processing capabilities. Furthermore, the next steps will involve implementing this advanced denoising technique into an early-exiting architecture to further enhance efficiency and effectiveness, particularly in real-time processing scenarios. These achievements underscore the potential of sophisticated signal processing techniques to enhance the reliability and performance of communication systems in diverse and noise-prone operational settings.

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